

Algebraic Geometry – Exercise Sheet 1

Week 1: Affine algebraic sets, projective space and localisation
Monday and Thursday afternoon sessions; Lectures 1–5

Difficulty. ★ routine; ★★ synthesis; ★★★ challenge or extension.

Use. The sheet is deliberately longer than the exercise sessions. Cerulean exercises form the suggested route; green exercises are alternatives.

Hints. Read them progressively and stop as soon as one is enough to restart the problem.

Monday afternoon — material available: Lecture 1 only

Exercise 1. Three points in the affine plane

★ **CORE**

Let

$$S = \{(0, 0), (1, 0), (0, 1)\} \subset \mathbb{k}^2.$$

(i) Show that

$$\mathbf{I}(S) = (x, y) \cap (x - 1, y) \cap (x, y - 1).$$

(ii) Prove directly that

$$\mathbf{I}(S) = (xy, x(x - 1), y(y - 1)).$$

(iii) Show that evaluation at the three points induces an isomorphism

$$\mathbb{k}[x, y]/\mathbf{I}(S) \simeq \mathbb{k}^3.$$

Exercise 2. Two axes and their coordinate rings

★ **CORE**

In $\mathbb{k}^2$, set $X = \mathbf{V}(x)$ and $Y = \mathbf{V}(y)$.

(i) Show that

$$X \cup Y = \mathbf{V}(xy), \quad X \cap Y = \mathbf{V}(x, y).$$

(ii) Compute the coordinate rings of X , Y , $X \cup Y$, and $X \cap Y$.

(iii) Explain geometrically why $\mathbb{k}[x, y]/(xy)$ has zero divisors.

Exercise 3. The parabola and the cusp

★★ **CORE**

Consider

$$P = \mathbf{V}(y - x^2), \quad C = \mathbf{V}(y^2 - x^3) \subset \mathbb{k}^2.$$

(i) Show that $t \mapsto (t, t^2)$ defines an isomorphism $\mathbb{A}^1 \rightarrow P$.

(ii) Show that $t \mapsto (t^2, t^3)$ defines a morphism $\nu : \mathbb{A}^1 \rightarrow C$ which is bijective on the underlying sets of $\mathbb{k}$ -points.

(iii) Write the corresponding homomorphism of coordinate rings and prove that ν is not an isomorphism.

Exercise 4. Polynomial maps of the affine line

★ **CHOICE**

Show that morphisms of affine algebraic sets $\mathbb{A}_{\mathbb{k}}^1 \rightarrow \mathbb{A}_{\mathbb{k}}^1$ are exactly polynomial maps $t \mapsto f(t)$. Describe the coordinate-ring homomorphism and check explicitly how composition reverses arrows.

Thursday afternoon — material available: Lectures 1–4

Exercise 5. Radicals and reduced rings

★ **CORE**

Compute

$$\sqrt{(x^2, y^3)}, \quad \sqrt{(x^2 y^3)}$$

in $\mathbb{k}[x, y]$. Decide which of the rings

$$\mathbb{k}[x]/(x^2), \quad \mathbb{k}[x, y]/(xy), \quad \mathbb{k}[x, y]/(x^2, xy)$$

are reduced.

Hint 1 (conceptual). Recall that R/I is reduced if and only if I is a radical ideal.

Hint 2 (technical). For $\sqrt{(x^2, y^3)}$, first show that both x and y lie in the radical, then use the prime ideal (x, y) for the reverse inclusion.

Hint 3 (technical). Write x^2y^3 as a product of powers of its distinct irreducible factors. Which multiplicities disappear when one takes the radical? For the quotient rings, try to exhibit a non-zero nilpotent whenever the defining ideal is not radical.

Exercise 6. Points at infinity

★ **CORE**

In the affine chart $x_0 \neq 0$ of $\mathbb{P}_{\mathbb{k}}^2$, write $x = x_1/x_0$ and $y = x_2/x_0$.

- (i) Homogenise the parabola $y = x^2$ and determine its points at infinity.
- (ii) Homogenise the hyperbola $xy = 1$ and determine its points at infinity.
- (iii) Explain geometrically why the answers are different.

Hint 1 (technical). Homogenise to the largest total degree occurring in the affine equation. Points at infinity are then obtained by adding the equation $x_0 = 0$.

Hint 2 (conceptual). The equation left on the line at infinity is the highest-degree homogeneous part of the original affine equation. Compare its projective zeroes for the parabola and the hyperbola.

Exercise 7. The line through two projective points

★ **CHOICE**

Let $p = [a_0 : \cdots : a_n]$ and $q = [b_0 : \cdots : b_n]$ be distinct points of $\mathbb{P}_{\mathbb{k}}^n$. Show that the line through them is defined by the 3×3 minors of

$$\begin{pmatrix} a_0 & \cdots & a_n \\ b_0 & \cdots & b_n \\ x_0 & \cdots & x_n \end{pmatrix}.$$

Write the equation explicitly for $n = 2$.

Hint 1 (conceptual). Choose non-zero representatives $a, b, x \in \mathbb{k}^{n+1}$. The point $[x]$ lies on the line through $[a]$ and $[b]$ precisely when x belongs to the two-dimensional vector space spanned by a and b .

Hint 2 (technical). Translate the preceding condition into a rank condition on the displayed matrix. Recall how rank at most 2 is detected by minors.

Exercise 8. Points in $\mathbb{P}^2$ from minors

★★ **CHOICE**

Show that the 2×2 minors of

$$M = \begin{pmatrix} x & 0 \\ y & y \\ 0 & z \end{pmatrix}$$

define the three coordinate points in $\mathbb{P}^2$. For four lines $\ell_1, \dots, \ell_4$ with no three concurrent, show that the maximal minors of

$$\begin{pmatrix} \ell_1 & 0 & 0 \\ -\ell_2 & \ell_2 & 0 \\ 0 & -\ell_3 & \ell_3 \\ 0 & 0 & -\ell_4 \end{pmatrix}$$

define their six pairwise intersection points.

Suggested strategy. First determine the support by computing the maximal minors. Only afterwards check that the scheme structure at each point is reduced.

Hint 1 (technical). For the first matrix, compute the three minors explicitly. If $xy = xz = yz = 0$ at a projective point, how many coordinates can be non-zero?

Hint 2 (technical). For the second matrix, deleting one row gives a product of three of the four linear forms. Show that all maximal minors vanish exactly when at least two of the ℓ_i vanish.

Hint 3 (technical). Near an intersection point $\ell_i = \ell_j = 0$, the other two linear forms are units because no three lines are concurrent. Use two of the minors to identify the local ideal with (ℓ_i, ℓ_j) .

Exercise 9. Equations of the Segre and Veronese images

★★★ **OPTIONAL**

a) For

$$([s_0 : s_1], [t_0 : t_1]) \mapsto [s_0 t_0 : s_0 t_1 : s_1 t_0 : s_1 t_1],$$

compute the kernel of $z_{ij} \mapsto s_i t_j$ and identify the image in $\mathbb{P}^3$.

b) For the quadratic Veronese map $\mathbb{P}^2 \rightarrow \mathbb{P}^5$, arrange the target coordinates as a symmetric 3×3 matrix and show that the image is defined by its 2×2 minors.

Suggested strategy. In each case, first find evident determinantal relations. The substantial point is then to prove that these relations generate the whole kernel, not merely that they vanish on the image.

Hint 1 (technical). For the Segre map, two monomials have the same image when their 2×2 exponent tables have the same row and column sums. Try to connect such tables by the elementary move corresponding to $z_{00}z_{11} = z_{01}z_{10}$.

Hint 2 (technical). For the Veronese map, view a monomial in the z_{ij} as a pairing of a multiset of indices. A 2×2 minor changes two pairs $(i, j), (k, \ell)$ into $(i, k), (j, \ell)$.

Hint 3 (technical). To prove that all pairings are connected by these switches, force one chosen pair to agree with the target pairing and argue by induction on the number of remaining pairs.

Exercise 10. Localising the coordinate axes

★ **CORE**

Let

$$A = \mathbb{k}[x, y]/(xy).$$

- (i) Compute A_x and A_y .
- (ii) Compute $A_{(x)}$ and $A_{(y)}$.
- (iii) Explain what these localisations retain and what they discard geometrically.

Hint 1 (conceptual). Keep separate the principal localisation A_x , which describes the open set $D(x)$, and the local ring $A_{(x)}$, which localises at the prime (x) .

Hint 2 (technical). In A_x , the element x is invertible, so the relation $xy = 0$ forces $y = 0$. For $A_{(x)}$, every non-zero polynomial in y lies outside (x) and becomes invertible.

Hint 3 (conceptual). The prime (x) is the generic point of the component $V(x)$, which is the y -axis. This explains why its local ring has fraction field $\mathbb{k}(y)$ rather than $\mathbb{k}(x)$.

Exercise 11. Localisation with a nilpotent

★ **CHOICE**

Let $A = \mathbb{k}[\varepsilon]/(\varepsilon^2)$. Compute

$$A_\varepsilon, \quad A_{(\varepsilon)}, \quad A_{1+\varepsilon}.$$

Explain why the three answers are so different.

Hint 1 (technical). The multiplicative set used to form A_ε contains $\varepsilon^2 = 0$. What happens to a localisation in which 0 becomes invertible?

Hint 2 (technical). Describe the elements outside the prime (ε) . Show directly that each $a + b\varepsilon$ with $a \neq 0$ is already a unit.

Hint 3 (technical). Compute $(1 + \varepsilon)(1 - \varepsilon)$.

Exercise 12. Two finite module structures

★★ **CHOICE**

Let

$$B = \mathbb{k}[x, y]/(y^2 - x).$$

Show that B is free of rank 2 over $\mathbb{k}[x]$ and free of rank 1 over $\mathbb{k}[y]$. Compare geometrically the corresponding morphisms to the affine line.

Hint 1 (technical). Use $x = y^2$. Every polynomial in y has a unique even-odd decomposition

$$f(y) = a(y^2) + yb(y^2).$$

This should produce a basis over $\mathbb{k}[x]$.

Hint 2 (conceptual). The two algebra structures give two different maps of the same parabola to an affine line: projection to the x -coordinate and parametrisation by the y -coordinate.

Exercise 13. Noether normalisation for the hyperbola

★★★ **CHOICE**

Let

$$B = \mathbb{k}[x, y]/(xy - 1), \quad u = x + y \in B.$$

Show that B is finite over $\mathbb{k}[u]$. Deduce that the corresponding morphism to $\mathbb{A}_{\mathbb{k}}^1$ has fibres with at most two points.

Suggested strategy. Replace the hyperbola by the Laurent polynomial ring $\mathbb{k}[x, x^{-1}]$, express u there, and look for a monic equation satisfied by x over $\mathbb{k}[u]$.

Hint 1 (technical). Since $y = x^{-1}$ and $u = x + x^{-1}$, multiply by x to obtain a quadratic equation for x . Also express x^{-1} in terms of u and x .

Hint 2 (technical). For the fibre over $u = a$, impose the relation $u - a = 0$. The resulting coordinate ring is governed by a polynomial of degree 2 in x .

After Friday's lecture or homework

Exercise 14. A presheaf which is not a sheaf

★ OPTIONAL

Let $X = U \amalg V$ be the disjoint union of two non-empty open subsets. Define a presheaf $\mathcal{F}$ by

$$\mathcal{F}(W) = \mathbb{Z} \quad \text{for } W \neq \emptyset, \quad \mathcal{F}(\emptyset) = 0,$$

with the evident restriction maps. Show that $\mathcal{F}$ is not a sheaf.

Hint 1 (conceptual). Use the cover $X = U \cup V$. Sections over U and V automatically agree on the empty intersection; choose two that cannot be restrictions of one integer over X .

Exercise 15. Continuous and holomorphic functions

★★ CHOICE

Let $U \subset \mathbb{C}$ be connected and contain at least two points. Show that $\mathcal{O}(U)$ has no zero divisors, whereas $\mathcal{C}^0(U)$ has non-zero functions f, g with $fg = 0$.

Hint 1 (conceptual). For holomorphic functions, think of the identity theorem: if one factor is not identically zero, the other vanishes on a non-empty open subset.

Hint 2 (technical). For continuous functions, choose two disjoint small closed discs in U and non-zero continuous functions supported in the two discs.

Exercise 16. $\mathcal{O}(d)$ on the projective line

★★★ OPTIONAL

Identify $\mathbb{P}_{\mathbb{C}}^1$ with $\mathbb{C} \cup \{\infty\}$. Define $\mathcal{O}(d)$ as the sheaf of meromorphic functions which are holomorphic away from ∞ and have at most a pole of order d there when $d \geq 0$; for $d < 0$, require a zero of order at least $-d$ at ∞ . Compute its global sections.

Suggested strategy. Work first on the affine chart $\mathbb{C} = \mathbb{P}^1 \setminus \{\infty\}$, then translate the condition at infinity using the coordinate $w = 1/z$.

Hint 1 (technical). A meromorphic function on $\mathbb{P}^1$ with no finite pole is a polynomial in z . A polynomial of degree m has a pole of order m at infinity.

Hint 2 (technical). When $d < 0$, a section is holomorphic everywhere and must vanish at infinity. Recall the global holomorphic functions on $\mathbb{P}^1$.